ZS-F27
The Register Fourier–Langlands Structure: The Seam Factorization $J = F^2S$, the Depth-One Obstruction, and the Three-Component Decomposition of the Adelic Locator

Kenny Kang
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Verification: 42/42 PASS (script v1.3; v1.0 correction of record in §10) | Zero Free Parameters | NO CLAIM on RH / GRH | Gate D5 remains IMPORTED-OPEN ≡ RH

§0. Abstract

The ZS-F24 → ZS-F25 → ZS-F26 arc reduced the corpus engagement with the Riemann spectrum to a sharply bounded honest terminus: the i-tetration pulsation is the archimedean scaling detector (PROVEN), the adelic locator is OPEN at the external frontier, and the single remaining statement of the Weil-positivity programme is the inductive rung whose totality is RH. The present paper adds one internal layer and one external layer to that terminus, both with zero free parameters. **Internally**, we prove that the corpus register $\mathbb{C}^{11}$ carries an exact, previously unregistered Fourier–Langlands structure. The discrete Fourier operator F on $\mathbb{C}^{11}$ satisfies the Seam Factorization $J = F^2S = S^{-1}F^2$ (Theorem 3.3, PROVEN): the corpus seam involution J is the parity square of F composed with the unit shift, and J is the unique parity-translate inverting the register position operator $D = \mathrm{diag}(j - 5)$ (Theorem 3.4, the Canonical Seam–Fourier Theorem, stated and proved for every odd N). The corpus transfer family $W_p = \mathrm{diag}(e^{2\pi i(j-5)/p})$ is shown to be the rational-time sampling $W_p = U(1/p)$ of the single flow $U(t) = e^{2\pi itD}$, with $\mathrm{order}(W_p) = p$ exactly; its multiplicative closure is $\bigoplus_p \mathbb{Z}/p$ with Pontryagin dual $\prod_p \mathbb{F}_p$ (Theorem 4.2, PROVEN). This yields the Depth-One Obstruction (Theorem 5.1, DERIVED): the register Wilson algebra realizes every prime at arithmetic depth exactly one, while the locator target requires the full prime-power towers $\{k \log p\}$ with $\zeta_p(k)$ weights (ZS-F25 Theorems 6.4-N/6.4-W) — the deficit is precisely the $k \geq 2$ rungs of the ZS-F26 window ladder. The register trace formula $\mathrm{Tr}\, U(t) = \sin(11\pi t)/\sin(\pi t)$ (order-11 Dirichlet kernel, PROVEN) has finite integer Fourier support and zero arithmetic singular support, confirming detector status at trace-formula level. The Fourier conjugate family $W_p^\vee = FW_pF^{-1}$ is circulant; the pair $(W_p, W_p^\vee)$ is a non-commuting Weyl-type clock–shift pair, and F intertwines the diagonal (Wilson) and circulant (convolution) algebras — the zero-dimensional instance of GL(1) geometric Langlands duality. Combining $J = F^2S$ with the ZS-M25 PROVEN identity $L_{1-s} = J\cdot L_s^\dagger\cdot J$ exhibits the corpus functional-equation involution as the square of the register Fourier operator: a finite-dimensional shadow of the Tate mechanism (DERIVED-interpretation). A negative result is registered: J is a charge-conjugation-type involution ($JW_pJ = W_p^*$), not an S-duality; the duality operator of the register is F. Version 1.1 adds two elementary sharpenings and one correction of record: the seam theorem is extended to every odd N; the DFT spectral multiset (3, 3, 3, 2) is shown to occur at N = 11 and at no other N (Proposition 3.7, with the dim Z link held at OBSERVATION); the freeness of the logarithmic prime sampling is upgraded to an exact lemma via unique factorization and Lindemann's theorem (Lemma 5.4); and the v1.0

verification-script incident (released script 27/28 against a stated 28/28, found by independent audit) is documented in §10. Version 1.2, prompted by a third audit, opens the seam-constrained finite Fourier programme with the paper's first theorems addressed to external mathematics, stated and proved without Z-Spin vocabulary: the Seam-Odd Uncertainty Theorem on prime cyclic groups — for nonzero f with $J_p f = -f$, $|\mathrm{supp}\, f| \cdot |\mathrm{supp}\, \hat{f}| \geq 2(p-1)$, a doubling of the prime Donoho–Stark bound, with complete equality classification (Theorem 3.9, via Tao's prime uncertainty principle); the Finite Trace Rigidity Theorem — no finite-dimensional self-adjoint flow, under any time sampling, can produce a Guinand–Weil-type logarithmic singular support (Theorem 5.5), of which Proposition 4.5 and the Depth-One Obstruction become instances; the half-angle phase pinning of seam-invariant mask polynomials (Lemma 3.11), seeding a seam-restricted Fuglede programme designated ZS-F28; and the generation identity $S = F^2 J$ — the seam and the Fourier transform alone generate the register's entire Weyl kinematics (Proposition 6.6). Version 1.3 adds the pilot (calibration) closure of that programme, with scope fixed by a fourth audit: on $\mathbb{Z}_{p^n q}$ the seam-invariant Fuglede equivalence is the restriction of the imported Malikiosis–Kolountzakis theorem (Corollary 3.12 — DERIVED, explicitly not a new theorem); seam-invariance is stable under every quotient (Lemma 3.13, the structural prerequisite of the F28 induction); non-vacuity is established by exhaustive enumeration at $N = 15$ (255 non-empty seam-invariant subsets; 16 spectral = 16 tiling, coinciding set-for-set, with non-coset examples); and the ZS-F28 main target — $N = p^n q^m$, seam-invariant spectral ⇒ tile — is formally designated with a five-step skeleton whose open core is a Lam–Leung-type Key Lemma. **Externally**, we import the 2024 closure of the unramified geometric Langlands conjecture (Gaitsgory–Raskin et al., GLC I–V), V. Lafforgue's function-field parameterization, the Fargues–Scholze geometrization of local Langlands, and the Etingof–Frenkel–Kazhdan analytic (self-adjoint) Langlands results, and prove the Three-Component Decomposition (Theorem 8.2, DERIVED-CONDITIONAL under the explicitly stated Langlands-reading Condition L): the adelic-locator OPEN decomposes as (A) the global curve over $\mathbb{Q}$ — OPEN at the frontier; (B) the non-abelian spectral correspondence — PROVEN wherever a curve exists, closed externally in 2024; (C) the unitarization layer — partial (EFK proven cases). The locator OPEN is thereby relocated and compressed, not closed; component (B), closed externally in 2024 but never registered by the ZS-F24 → ZS-F26 arc, which did not engage the Langlands column, enters the corpus frame here as a theorem of the external frontier. Table 7.1 of ZS-F25 is extended by two rows, the EFK compactness mechanism is registered as a closure-path template for O-F26.5, and gate FM13-5 is reframed exactly as the gap between the harmonic embedding $p \mapsto 1/p$ (torsion, register side) and the logarithmic embedding $p \mapsto \log p$ (free, locator side) of the primes into one and the same flow. No claim is made on RH, on GRH, or on the construction of the missing curve. Genre statement (§1.1): this is an operator-normalization and registration paper; its mathematical constituents are standard finite Fourier analysis, and its principal contribution is corpus-internal.

Keywords: register Fourier operator, seam factorization, discrete Fourier transform, finite Fourier–Mukai duality, Satake parameter, depth-one obstruction, Dirichlet kernel, Weyl clock–shift pair, Tate shadow, geometric Langlands, Gaitsgory–Raskin, Fargues–Scholze, analytic Langlands, adelic locator, Weil positivity, Z-Spin mediation, anti-numerology.

§0.1 Epistemic Status Legend

TAG	MEANING

PROVEN	Rigorous mathematical proof or exact closed-form identity under declared definitions; verified by direct computation; no open step.
DERIVED	Follows by valid deduction from corpus axioms (A, Q, dim Z) and PROVEN results; zero new free parameters.
DERIVED-CONDITIONAL	Follows given one explicitly stated, currently open, condition (here: Condition L of §8).
DERIVED-interpretation	Synthesis reading of PROVEN/DERIVED items as a unified pattern; no new theorem asserted.
IMPORTED-PROVEN	Result proved externally (peer-reviewed or community-verified frontier) and used without re-proof; full citation given.
IMPORTED (partial)	External result proven in stated special cases and open in general; consumed only at its proven extent.
IMPORTED-OPEN	Unresolved statement of the external frontier, imported at its registered status (e.g. RH, Weil positivity); not a corpus hypothesis.
COMPUTED	Numerical result under the stated discretization; consistency evidence, never proof.
OBSERVATION	Exact structural fact registered without a mechanism; carries no evidentiary weight in proofs.
HYPOTHESIS	Motivated conjecture with explicit closure path; not derived.
OPEN	Registered open gate; not closed by present corpus tools.
NON-CLAIM	Stated explicitly as not asserted; documented to prevent overclaim.
LOCKED	Prior Z-Spin value or operator used without modification.

§1. Introduction and Position in the Programme

The RH-bridge arc of the corpus has a deliberately conservative shape. ZS-F24 retired the strongest internal conjecture — that the i-tetration pulsation might itself be the Riemann locator — by identifying the pulsation with the archimedean scaling (detector) piece of the Connes prolate operator and registering the adelic locator as OPEN at the live external frontier. ZS-F25 made the exclusion unconditional (the rank-one length spectrum of the pulsation against the infinite-rank prime-logarithm spectrum required by the Guinand–Weil explicit formula), proved the Place-Selection Theorem (the Z-sector substrate is ℂ, the archimedean place, and every p-adic substrate is excluded), and closed the logical perimeter of the Weil-positivity programme around Gate D5. ZS-F26 converted the remaining audit items into theorems — the unconditional GNS core, the Douglas uniformization of the rung, the Master Inequality MI(2) — and left the inductive rung OPEN, its totality being RH.

Two questions arose in the internal deep-exploration sessions that prepared this paper, one internal and one external. The internal question: the corpus has, since ZS-M4 and ZS-M25, carried a commuting family of prime-indexed unitaries W_p on the register $\mathbb{C}^{11}$ and a seam involution J satisfying $J·W_p·J = W_p^*$ and $L_{1−s} = J·L_s^\dagger·J$ (both PROVEN) — but it has never asked what duality operator generates these identities. Is there a canonical Fourier–Langlands structure on the register itself, of which J and the W_p are shadows? The external question: between the writing of ZS-F24 (which registered the locator OPEN) and the present paper, the external frontier changed — the unramified geometric Langlands conjecture was proved in 2024 (Gaitsgory–Raskin et al.). Does that closure alter the shape of the locator OPEN, and if so, how exactly?

This paper answers both with zero free parameters. The internal answer is a small but exact body of theorems: the register discrete Fourier operator F factors the seam involution ($J = F^2 S$, §3), organizes the entire W_p family as the rational-time sampling of one flow (§4), yields a new obstruction theorem complementing the ZS-F25 rank obstruction (the Depth-One Obstruction, §5), produces the dual ('t Hooft-type) family and the register Weyl kinematics (§6), and — combined with the ZS-M25 identity — exhibits the corpus functional-equation involution as the square of a Fourier operator, a finite shadow of the Tate mechanism (§7). The external answer is a registration theorem: under one explicitly stated condition, the locator OPEN decomposes into three components, of which the middle one — a conjecture of three decades' standing until its external closure in 2024, a closure the ZS-F24 → ZS-F26 arc never registered because that arc did not engage the Langlands column — is here registered as externally PROVEN wherever its geometric prerequisite (a curve) exists (§8). The two answers are organically linked: the register structure of §3–§7 is the GL(1), depth-one, torsion shadow of exactly the machinery whose non-abelian completion §8 locates at the frontier.

Nothing in this paper claims RH, constructs the missing global curve, or asserts that geometric Langlands applies to $\mathbb{Q}$ today. The paper's function is the corpus's standard one at this boundary: to prove what is provable, to relocate what cannot yet be closed, and to pre-register the tripwires that would otherwise invite overclaim.

§1.1 Genre and the external-novelty disclaimer

Genre statement (added in v1.1, prompted by external audit). This is an operator-normalization and registration paper. Its mathematical constituents — the fourth-root and parity properties of the DFT, the diagonal–circulant intertwining, finite Weyl pairs, Pontryagin duality of $\bigoplus_p \mathbb{Z}/p$, the Dirichlet kernel — are standard finite Fourier analysis, with primary references dating to 1962–1981 [1, 12, 13]; the paper claims no new external mathematics beyond two elementary sharpenings (Theorem 3.4, stated for every odd N, and Proposition 3.7, the $N = 11$ multiset uniqueness) and one exact lemma assembled from imported theorems (Lemma 5.4). The principal contribution is corpus-internal: the identification of the LOCKED operators J and W_p as projections of one canonical Fourier package — so that their definitions are exhibited as forced, not chosen — the second no-go (the Depth-One Obstruction), the trace-level detector confirmation, and the frontier registration of §8–§9. The three-component narrative of §8 is, as its provenance note records, the consensus articulation of the cited external programs; the paper contains no external synthesis export analogous to the (H, h, P) triage of ZS-F25, and its self-description should be read accordingly: an internal floor is proved, and an external ceiling is registered — not contributed to. Version 1.2 amends this statement in one respect, in the direction the audit prescribed (external mathematics first, corpus reinterpretation after): §3.5–§3.6, §5.4, and §6.5 now contain theorems addressed to external finite Fourier analysis, stated and proved Z-Spin-free, with corpus readings appended only after the proofs; the paper remains an operator-normalization paper, now seeding the seam-constrained programme designated ZS-F28.

§2. Locked Inputs and Imports

ZS-F27 is a math-spine paper. No proof below consumes the numerical values of **A** = 35/437 or z^* = 0.438283 + 0.360592i; the corpus constants enter only through the LOCKED register dimension **Q** = 11 and the LOCKED operator definitions. Zero new free parameters are introduced.

Table 2.1. Locked inputs (internal) and imports (external).

Object	Statement	Status
Q = 11 register	State space $\mathbb{C}^{11}$, basis $\|0\rangle \dots \|10\rangle$ (ZS-F5).	LOCKED
Seam involution J	$J\|j\rangle = \|10 - j\rangle$ (ZS-F5, ZS-M22).	LOCKED
Transfer family W_p	W_p = diag(e^{2πi(j−5)/p}) (ZS-M4).	LOCKED
J·W_p·J = W_p*	ZS-M25 Appendix B, Category F.	IMPORTED-PROVEN
L_{1−s} = J·L_s†·J	ZS-M25 Appendix B, Category F (Berry–Keating family).	IMPORTED-PROVEN
Guinand–Weil support necessity	ZS-F25 Theorem 6.4-N: locator support = {±k log p}, all prime powers.	IMPORTED-PROVEN
Weight necessity ζ_p(k)	ZS-F25 Lemma 6.4-W: Selberg/Weil weight ratio = (1 − p^{−k})^{−1}.	IMPORTED-PROVEN
Rank obstruction	ZS-F25 Lemma 6.5 / Corollary 6.6: pulsation length-rank 1 vs required rank ∞.	IMPORTED-PROVEN
Classification Table 7.1	ZS-F25: invariants (H, h, P); empty quadrant (non-abelian, h > 0, adelic).	IMPORTED
F26 window ladder	ZS-F26 §5–§7: rungs at prime powers k log p; MI(2); O-F26.5.	IMPORTED
Z = ∂X	Bedrock postulate, corpus-wide conditional (O-Q16.12).	IMPORTED
Geometric Langlands (unramified)	Gaitsgory–Raskin et al., GLC I–V (2024): D-mod(Bun_G) ≃ IndCoh_N(LocSys_Ĝ) over a smooth projective complex curve [3].	IMPORTED-PROVEN
Function-field parameterization	V. Lafforgue (2018): automorphic → Galois via excursion operators, any reductive G over function fields [4].	IMPORTED-PROVEN
Local geometrization	Fargues–Scholze (2021): local Langlands as geometric Langlands on the Fargues–Fontaine curve, one prime p [5].	IMPORTED-PROVEN
Analytic Langlands	Etingof–Frenkel–Kazhdan (2019–2023): Hecke operators on L²(Bun_G) compact, self-adjoint, commuting; real discrete spectrum in proven (genus-0, parabolic) cases [6].	IMPORTED (partial)
Tate mechanism	Tate (1950): functional equation of ζ from Fourier duality on the adeles [2].	IMPORTED-PROVEN
DFT eigenstructure	McClellan–Parks (1972): $F^4 = I$; eigenvalue multiplicities of the DFT [1].	IMPORTED-PROVEN
Pontryagin duality	Dual of ⊕_p ℤ/p is ∏_p ℤ/p; dual of ℚ/ℤ is $\hat{\mathbb{Z}}$ [13].	IMPORTED-PROVEN
Lindemann transcendence	e^a is transcendental for every nonzero algebraic a; consumed in Lemma 5.4 [16].	IMPORTED-PROVEN
Tao prime uncertainty	For nonzero f on ℤ_p (p prime): |supp f| + |supp $\hat{f}$| ≥ p + 1; consumed in Theorem 3.9 [17].	IMPORTED-PROVEN

Fuglede on $\mathbb{Z}_{p^n q}$	Spectral ⇔ tile on cyclic groups of order $p^n q$; consumed in Corollary 3.12 [22].	IMPORTED-PROVEN

§3. The Register Fourier Operator and the Seam Factorization

§3.1 Definitions and conventions

Fix $\omega = e^{2\pi i/11}$ and define on $\mathbb{C}^{11}$ the unitary discrete Fourier operator F, the cyclic shift S, and the register position operator **D**:

$$F|j\rangle = (1/\sqrt{11}) \Sigma_k \omega^{jk} |k\rangle, \qquad S|j\rangle = |j+1 \bmod 11\rangle, \qquad D = \mathrm{diag}(j-5),\ j = 0, \ldots, 10.$$

F is the canonical Fourier–Mukai duality operator of the finite abelian group $\mathbb{Z}/11$ — the zero-dimensional instance of GL(1) geometric Langlands duality [12, 13] — and D is the position operator centered at the kinematic fixed-point slot $|5\rangle$ (ZS-F11), with spectrum $\{-5, \ldots, 5\}$. The convention ω versus $\bar{\omega}$ is fixed once here; the single convention-dependent statement of this paper is flagged in §10.

§3.2 The factorization theorems

Theorem 3.2 (Fourier square = parity). [PROVEN] F is unitary, $F^4 = I$, and $F^2 = P$, where $P|j\rangle = |-j \bmod 11\rangle$ is the parity operator. *Proof.* $F^2|j\rangle = (1/11) \Sigma_{k,l} \omega^{k(j+l)} |l\rangle = \Sigma_l \delta_{j+l \equiv 0\ (11)} |l\rangle = |-j \bmod 11\rangle$. Hence $F^4 = P^2 = I$. □ (Verified: checks B1–B3.)

Theorem 3.3 (Seam Factorization). [PROVEN] The corpus seam involution satisfies

$$J = F^2S = S^{-1}F^2, \qquad \text{and} \qquad F^2 S F^2 = S^{-1}.$$

Proof. $F^2S|j\rangle = F^2|j+1\rangle = |-(j+1) \bmod 11\rangle = |10-j\rangle = J|j\rangle$ for every $j = 0, \ldots, 10$. The second form follows since parity conjugates the shift to its inverse: $F^2SF^2 = PSP = S^{-1}$, so $S^{-1}F^2 = F^2S$. Involutivity is then structural: $J^2 = F^2SF^2S = S^{-1}S = I$. □ (Verified: checks C1–C3, A2.)

The seam involution — the central duality of the register, generator of the D_4 register symmetry (ZS-F0 §8.13) and implementer of the corpus functional-equation symmetry (ZS-M25) — is therefore not an independent primitive: it factors exactly through the square of the register Fourier operator. F is a canonical fourth root of the identity whose square is J shifted; equivalently, F is a square root of the operator $J \cdot S^{-1}$.

Theorem 3.4 (Canonical Seam–Fourier Theorem). [PROVEN] Let N be odd, let F and S be the DFT and cyclic shift on $\mathbb{C}^N$, and let $D = \mathrm{diag}(j - (N-1)/2)$ be the centered register position operator. Among the N parity-translates $T_a = F^2S^a$ ($a = 0, \ldots, N-1$), exactly one reverses the register orientation: $T_a D T_a^{-1} = -D$ if and only if $a = 1$. The unique reversing translate $J_N := F^2S$ is an involution, and at $N = 11$ it coincides with the corpus seam involution: $J_{11} = J$. *Proof.* $T_a|j\rangle = |-j-a \bmod N\rangle$, so $T_a D T_a^{-1}$ has eigenvalue $((-k-a \bmod N) - (N-1)/2)$ at $|k\rangle$; equality with $-(k - (N-1)/2)$ for all k forces $-k-a \equiv (N-1)-k \pmod N$, i.e. $a \equiv 1 \pmod N$. Involutivity: $J_N^2 = F^2SF^2S = S^{-1}S = I$. The $N = 11$ identification is

Theorem 3.3. □ (Verified: check C4 at $N = 11$; check C6 across odd $N \in \{5, 7, 9, 11, 13\}$.) The corpus seam is therefore not an arbitrary reflection: it is the canonical centered Fourier-parity shift reversing the register orientation — unique at every odd register size — and the corpus-specific content is the identification $J = J_{11}$. (Universality flag: the theorem holds for every odd N and does not select $Q = 11$; see §3.4 and §10. Promoted from a verification check of v1.0 to a main theorem at the suggestion of external audit.)

Corollary 3.5 (re-derivation of the ZS-M25 identities). [PROVEN] $JDJ = -D$ (check C5); consequently $J e^{2\pi itD} J = e^{-2\pi itD}$ for all real t, which at $t = 1/p$ reproduces the ZS-M25 PROVEN identity $J\cdot W_p\cdot J = W_p^*$ (check D1). The seam factorization thus supplies the structural origin of the M25 Category-F identities: they are the t-flow image of the single relation $JDJ = -D$, itself forced by Theorem 3.4.

Remark 3.6 (universality flag). F^2 = parity holds for the DFT on $\mathbb{C}^N$ for every N (verified additionally at $N = 7, 12$; check I2). Theorem 3.3 therefore does not select $Q = 11$; its content is the exact identification of the corpus-specific J with the canonical operator F^2S, not a uniqueness claim about 11. This flag is part of the anti-numerology audit (§10).

§3.4 The spectral multiset of F and the uniqueness of N = 11

Proposition 3.7 (multiset uniqueness). [PROVEN] By the McClellan–Parks multiplicity count [1], the eigenvalue multiplicities of the DFT on $\mathbb{C}^N$ over $\{1, -1, i, -i\}$ form, as a multiset, $\{m+1, m, m, m-1\}$ for $N = 4m$, $\{m+1, m, m, m\}$ for $N = 4m+1$, $\{m+1, m+1, m, m\}$ for $N = 4m+2$, and $\{m+1, m+1, m+1, m\}$ for $N = 4m+3$. Consequently the spectral multiset $\{3, 3, 3, 2\}$ occurs if and only if $N = 4m+3$ with $m = 2$, i.e. if and only if $N = 11$. (Scan witness over $2 \leq N \leq 60$: check H2.) Under the $\omega = e^{2\pi i/11}$ convention of §3.1 the deficient two-dimensional eigenspace is $\ker(F + iI)$; under the conjugate convention it is $\ker(F - iI)$; the convention-invariant statement is that exactly one chirality $\pm i$ is deficient and its eigenspace has dimension 2.

Guard. [OBSERVATION + NON-CLAIM] The coincidence dim(deficient $\pm i$ eigenspace) = 2 = dim Z — the register Fourier operator possessing exactly one deficient chiral eigenspace, of the Z-sector dimension, at exactly the corpus register size — is registered as OBSERVATION only. Proposition 3.7 does not derive $Q = 11$: the multiset criterion is a single scalar condition, and other patterns single out other N (e.g. $\{3, 2, 2, 2\}$ singles out $N = 9$); the statement bears on the corpus only jointly with the ZS-F5 slot decomposition $(Z, X, Y) = (2, 3, 6)$, which fixes $Q = 11$ independently. The observation enters no proof in this paper.

§3.5 Seam-odd uncertainty on prime cyclic groups

This subsection and the next open the seam-constrained finite Fourier programme proposed by the third audit: instead of reinterpreting standard DFT facts, impose the canonical seam $J_N = F^2S$ as a symmetry constraint and prove what it buys. Both are stated and proved for plain cyclic groups, with no Z-Spin vocabulary; the corpus reading is appended after the proofs.

Lemma 3.8 (seam-odd structure). [PROVEN] Let N be odd, $f \in \mathbb{C}^N$ nonzero with $J_N f = -f$ (seam-odd; here $(J_N f)(k) = f(N-1-k)$). Then: (i) $f((N-1)/2) = 0$, supp f is a disjoint union of seam pairs $\{k, N-1-k\}$, and $|\text{supp } f|$ is even, ≥ 2; (ii) under the §3.1 convention, $\hat{f}(-m) = -\omega^m \hat{f}(m)$ for all m; hence $\hat{f}(0) = 0$, supp $\hat{f}$ is paired under $m \mapsto -m$, and $|\text{supp } \hat{f}|$ is even with supp $\hat{f} \subseteq \mathbb{Z}_N \setminus \{0\}$. *Proof.* (i) Immediate from $f(N-1-k) = -f(k)$: the center is the unique fixed point of the seam reflection. (ii) Substituting $k \mapsto N-1-k$ in $\hat{f}(-m) = \Sigma_k f(k)\omega^{-km}$ and using $-(N-1-k) \equiv k+1 \pmod N$ gives $\hat{f}(-m) = -\omega^m \hat{f}(m)$; at $m = 0$ this forces $\hat{f}(0) = 0$, and for odd N the involution $m \mapsto -m$ has 0 as its only fixed point. □ (Verified: check U1.)

Theorem 3.9 (Seam-Odd Uncertainty Theorem). [PROVEN] Let p be an odd prime and $f \in \mathbb{C}^p$ nonzero with $J_p f = -f$. Then

$$|\text{supp } f| \cdot |\text{supp } \hat{f}| \;\geq\; 2(p-1)\,,$$

with equality if and only if f is a seam dipole $f = c(\delta_k - \delta_{p-1-k})$, $k \neq (p-1)/2$, or $\hat{f}$ is a dual dipole supported on $\{m, -m\}$, $m \neq 0$, with $\hat{f}(-m) = -\omega^m \hat{f}(m)$ — in the latter case $f(k) = c\cdot\omega^{-km}(1 - \omega^{(2k+1)m})$, a discrete wave vanishing exactly at the center slot. *Proof.* Write $a = |\text{supp } f|$, $b = |\text{supp } \hat{f}|$. By Lemma 3.8, a and b are even, $a \geq 2$, and (center, respectively 0, excluded) $a \leq p-1$, $b \leq p-1$. By Tao's uncertainty principle for prime cyclic groups [17], $a + b \geq p + 1$. For fixed even a the minimum admissible b is $\max(2, p+1-a)$, so $ab \geq a(p+1-a)$, a concave quadratic in a minimized on the even range $[2, p-1]$ at its endpoints, where it equals $2(p-1)$; interior even values $a \geq 4$ give $ab \geq 4(p-3) > 2(p-1)$ for $p > 5$, and the cases $p = 3, 5$ are checked directly. Hence $ab \geq 2(p-1)$, and equality forces $\{a, b\} = \{2, p-1\}$. If $a = 2$, the pairing of Lemma 3.8(i) forces the dipole form, and $\hat{f}(m) = c\omega^{km}(1 - \omega^{(p-1-2k)m})$ vanishes iff $(p-1-2k)m \equiv 0 \pmod p$, i.e. iff $m = 0$ since $p-1-2k \not\equiv 0$; so $b = p-1$ exactly. If $b = 2$, Lemma 3.8(ii) forces the dual-dipole form, and the displayed f vanishes iff $(2k+1)m \equiv 0 \pmod p$, i.e. only at $k = (p-1)/2$; so $a = p-1$ exactly. □ (Verified: checks U2–U3, with the minimum attained exactly at $2(p-1)$ for $p \in \{5, 7, 11\}$.)

Remark 3.10 (calibration and scope). The unconstrained prime benchmark is $|\text{supp } f|\cdot|\text{supp } \hat{f}| \geq p$ (Donoho–Stark [18], or Tao [17] via $ab \geq a(p+1-a) \geq p$ at $a = 1$): the seam-odd constraint roughly doubles the uncertainty product, and the theorem closes with a complete extremizer classification — the audit's Theorem candidate A, registered PROVEN. The composite-N extension (Tao unavailable; Meshulam-type divisor bounds [19] expected to control the answer) is O-F27.4. Corpus reading, appended after the proof as the programme prescribes: a seam-odd (C_-) channel on the register cannot localize jointly in slot and dual-slot beyond the doubled bound — the seam symmetry is an information constraint, not a label. No corpus statement is used in the proof.

§3.6 Seam-invariant masks: half-angle pinning and the ZS-F28 programme

Lemma 3.11 (half-angle phase pinning). [PROVEN] Let N be odd and $A \subseteq \mathbb{Z}_N$ with mask polynomial $A(x) = \Sigma_{a\in A} x^a$. Then A is seam-invariant ($A = N-1-A$) if and only if $\zeta^{-(N-1)/2} A(\zeta) \in \mathbb{R}$ for every N-th root of unity ζ; equivalently $A(x) \equiv x^{N-1}A(x^{-1}) \pmod{x^N - 1}$. *Proof.* Seam-invariance means the centered set $A - (N-1)/2$ is symmetric under negation, so its mask $\zeta^{-(N-1)/2}A(\zeta)$ is a sum of cosines, hence real; the converse and the polynomial form follow by reversing the computation, using that the coefficients are 0/1. □ (Verified: check P1 at $N = 11, 15$.) Consequence: at every cyclotomic point the Fourier phase of a seam-invariant mask collapses from a circle to a line, and every divisibility pattern $\Phi_d(x) \mid A(x)$ must be compatible with the pinning.

Programme registration (audit candidate B). [HYPOTHESIS-strong] The seam-restricted Fuglede programme — prove, for at least one nontrivial family of odd cyclic groups, that seam-invariant spectral sets tile (or the converse), by imposing Lemma 3.11's pinning inside the Coven–Meyerowitz cyclotomic framework [21], starting where the unrestricted conjecture is settled (e.g. $\mathbb{Z}_{p^n q}$ [22]; the conjecture is open in general [20–21]) — is registered as O-F27.5 and designated the ZS-F28 programme. Nothing of it is proved here; Lemma 3.11 is its only currently PROVEN ingredient, and the audit's assessment (the candidate of highest external value, at medium-to-high difficulty) is adopted as stated.

§3.7 Pilot closure of the seam-restricted programme (calibration, not theorem)

Scope, fixed by the fourth audit: this section carries only the pilot — the closure of the first imported base family and the structural lemma the F28 induction presupposes. The novelty-bearing target is separated into ZS-F28; attempting to close it here would change the paper's identity from register paper to Fuglede case-analysis paper. The prohibited phrasing "we close the seam-restricted Fuglede programme" is not asserted anywhere (NC-F27.7): what is closed below is an already-settled base family, and what is isolated is the mechanism F28 must prove.

Corollary 3.12 (pilot closure on $\mathbb{Z}_{p^n q}$). [DERIVED (from IMPORTED-PROVEN)] Let $N = p^n q$ be odd with distinct primes p, q, and let $A \subseteq \mathbb{Z}_N$ be seam-invariant. If A is spectral, then A tiles $\mathbb{Z}_N$; conversely, if A tiles, then A is spectral. *Proof.* Fuglede's conjecture holds on $\mathbb{Z}_{p^n q}$ in both directions [22]; the corollary is its restriction to the seam-invariant subclass. □ This is explicitly not a new theorem and contributes no new external mathematics: the unrestricted class is closed, hence every subclass is closed. Its role is calibration — the F28 programme operates on non-empty, already-closed terrain, and the seam constraint can now be studied against a known answer.

Lemma 3.13 (seam quotient stability). [PROVEN] Let N be odd and $M \mid N$. The reduction $\pi : \mathbb{Z}_N \to \mathbb{Z}_M$ intertwines the seam maps: $\pi(N-1-x) = M-1-\pi(x)$. Hence the image of a seam-invariant set is seam-invariant. *Proof.* The seam is $x \mapsto -1-x$, and -1 reduces to -1 under every quotient. □ (Verified: check Q1, all 256 seam-invariant subsets of $\mathbb{Z}_{15}$ under $\mathbb{Z}_{15} \to \mathbb{Z}_5, \mathbb{Z}_3$.) Significance: this is the structural prerequisite of the F28 induction (Step 4 of the skeleton below) — fiber-quotient descent stays inside the seam-invariant category. The audit's induction step presupposed it implicitly; it is supplied here as a theorem so that the programme's induction is well-posed before drafting begins.

Non-vacuity. [COMPUTED] Exhaustive enumeration at the smallest base case $N = 15 = pq$: of the 255 non-empty seam-invariant subsets, exactly 16 are spectral and exactly 16 tile, and the two classes coincide set-for-set (check S1) — the pilot equivalence holds with zero mismatches over the full subclass. The class is non-trivial: it contains non-coset tiles, e.g. $\{0, 7, 14\} = 7\cdot\{0, 1, 2\}$ (check S2). One forced feature, recorded in passing: every seam-invariant tile of an odd cyclic group contains the center slot — $|A|$ divides odd N, so $|A|$ is odd, and a seam-invariant set of odd cardinality must contain the seam's unique fixed point (trivially PROVEN); that at $N = 11$ this center is $|5\rangle$, the kinematic fixed point of ZS-F11, is registered as OBSERVATION only, used in no proof.

ZS-F28 main target (formal designation). [HYPOTHESIS-strong] $N = p^n q^m$, $p \neq q$ odd primes, $A \subseteq \mathbb{Z}_N$ seam-invariant: if A is spectral, then A tiles. F28's novelty claim is confined to this direction; the converse (tile ⇒ spectral) in this family is expected to follow from the Coven–Meyerowitz machinery (their T2 condition holds for tiles whose cardinality has at most two prime divisors [21]) together with Łaba's criterion (T1 and T2 imply spectral) [24] — an expectation to be confirmed against the literature at F28 drafting, recorded here for the audit trail. Five-step skeleton: (1) pass to the centered self-reciprocal mask (Lemma 3.11: $A_c(x^{-1}) = A_c(x)$, all cyclotomic values real); (2) convert the spectral condition into the cyclotomic divisor set $D(A) = \{d : \Phi_d \mid A(x)\}$ forced by the spectrum's difference set; (3) the Key Lemma — the open core: for mixed divisors $\Phi_{p^a q^b} \mid A(x)$, the Lam–Leung decomposition of vanishing sums of roots of unity into p-cycles and q-cycles [23], constrained by self-reciprocity, collapses mixed patterns into symmetric fibers; (4) induction on $n + m$ by fiber-quotient descent, well-posed by Lemma 3.13, with base case $\mathbb{Z}_{p^n q}$ = Corollary 3.12; (5) conclude tiling via the Coven–Meyerowitz conditions [21]. The Key Lemma's life or death is to be decided by structural enumeration (the reconnaissance step) before F28 drafting; if it dies, the registered fallback is a weakened hypothesis (e.g. assuming T2) and the obstruction itself becomes the registrable result.

§4. The Wilson Flow and Its Satake Reading

§4.1 One flow, all primes

Proposition 4.1 (single-flow membership). [PROVEN] Define the one-parameter unitary group $U(t) = e^{2\pi i t D}$. Then for every prime p,

$$W_p = U(1/p),$$

i.e. the entire corpus transfer family is the harmonic-time sampling of one flow generated by the register position operator **D**. (Verified: check D2.) The family is commuting (check D4) because it lies on one flow — the structural reason behind the diagonal form noted since ZS-M4.

Theorem 4.2 (order and closure). [PROVEN] (i) $\mathrm{order}(W_p) = p$ exactly. (ii) $U(t) = I$ if and only if $t \in \mathbb{Z}$, so $t \bmod 1 \mapsto U(t)$ is injective on $\mathbb{Q}/\mathbb{Z}$. (iii) The multiplicative closure of $\{W_p : p \text{ prime}\}$ is

$$\langle W_p\rangle_p = \{ U(t) : t \in \Sigma_p (1/p)\mathbb{Z} \bmod 1 \} \cong \oplus_p \mathbb{Z}/p,$$

one cyclic factor of order p per prime, with no element of order p^2 for any p. *Proof.* (i) $W_p^p = \mathrm{diag}(e^{2\pi i(j-5)}) = I$, and $W_p \neq I$ since $e^{2\pi i/p} \neq 1$ at $j = 6$. (ii) $U(t) = I$ iff $t(j - 5) \in \mathbb{Z}$ for all j, iff $t \in \mathbb{Z}$ (take $j = 6$). (iii) By (ii) the closure is isomorphic to the subgroup of $\mathbb{Q}/\mathbb{Z}$ generated by $\{1/p\}$, which is $\oplus_p (1/p)\mathbb{Z}/\mathbb{Z} \cong \oplus_p \mathbb{Z}/p$; its p-primary component is $\mathbb{Z}/p$ exactly, so $1/p^2$ (order p^2) is not attained. □ (Verified: checks D3, E1–E3, including $|\langle W_2, W_3, W_5\rangle| = 30$.)

Corollary 4.3 (Pontryagin dual). [PROVEN] The Pontryagin dual of the register Wilson group is $\prod_p \mathbb{Z}/p = \prod_p \mathbb{F}_p$ — one prime field per prime, each at depth one. By contrast the dual of the full $\mathbb{Q}/\mathbb{Z}$ is the profinite completion $\hat{Z} = \prod_p \mathbb{Z}_p = \prod_p \lim \mathbb{Z}/p^k$ — the full pro-p towers. The register family realizes exactly the depth-one floor of the adelic tower.

§4.2 The Satake reading

Proposition 4.4 (GL(1) Satake parameterization). [DERIVED-interpretation] The assignment $W_p \mapsto 1/p \in \mathbb{Q}/\mathbb{Z}$ — equivalently $W_p \mapsto \omega_p = e^{2\pi i/p} \in U(1)$, with $W_p = \omega_p^D$ — is a group isomorphism of the Wilson family onto its image and assigns to each prime a torsion point of the GL(1) dual torus. This is the register's unramified Satake parameterization at GL(1): one commuting Hecke-type operator per prime, one dual-torus parameter per prime. The parameters are torsion (roots of unity); a locator-grade family would require non-torsion parameters of modulus governed by p^{-s} — the torsion/non-torsion dichotomy is the parameter-level face of the detector/locator dichotomy. No theorem is asserted beyond the isomorphism (which is Theorem 4.2(iii)); the Satake naming is interpretive.

§4.3 The register trace formula is a Dirichlet kernel

Proposition 4.5. [PROVEN] For all real t,

$$\mathrm{Tr}\, U(t) \;=\; \sum_{j=0}^{10} e^{2\pi i(j-5)t} \;=\; \sin(11\pi t) / \sin(\pi t) \,,$$

the order-11 Dirichlet kernel. (Verified: check G1.) Its Fourier support is the finite integer set $\{-5, \ldots, 5\}$ (check G2): the register trace formula has no singular support at any $\pm k \log p$. By ZS-F25 Theorem 6.4-N — the spectral measure of any Riemann locator carries delta-type structure at every prime-power logarithm — this is the trace-formula-level confirmation, exact and parameter-free, that the register flow is pure detector. The pulsation was excluded in ZS-F25 by its rank-one length spectrum; the register flow is now excluded by its band-limited trace. The two exclusions are independent in mechanism and concordant in verdict.

§5. The Depth-One Obstruction

Theorem 5.1 (Depth-One Obstruction). [DERIVED] The register Wilson algebra realizes every prime at arithmetic depth exactly one: its multiplicative closure is $\bigoplus_p \mathbb{Z}/p$ (Theorem 4.2), whose p-primary component is $\mathbb{Z}/p$, while the locator target requires, by ZS-F25 Theorem 6.4-N, closed-orbit/support structure at every prime power $k \log p$ ($k = 1, 2, 3, \ldots$), i.e. the full towers whose inverse limits assemble $\hat{Z} = \prod_p \mathbb{Z}_p$; and requires, by ZS-F25 Lemma 6.4-W, orbit weights departing from the Selberg form by exactly the local factor

$$\zeta_p(k) \;=\; (1 - p^{-k})^{-1} \;=\; \sum_{m\geq 0} p^{-mk} \,,$$

the geometric series over depth. The deficit between the register family and the locator target is therefore precisely the $k \geq 2$ layers — in ZS-F26 ladder terms, every rung at a proper prime power (the first being $p^k = 4$ at window $2 \log 2$) is invisible to the register Wilson family, which can shadow only the depth-one rungs (primes p at log p). *Proof.* Immediate from Theorem 4.2(iii), Corollary 4.3, and the two imported necessities. □

Corollary 5.2 (two independent obstructions). [DERIVED] The corpus now holds two independent finite-versus-infinite exclusions of its internal structures from locator status: the length-rank obstruction (ZS-F25 Lemma 6.5: pulsation rank 1 vs required rank ∞, over the dynamics) and the depth-one obstruction (this paper: Wilson closure depth 1 vs required depth ∞, over the arithmetic tower). They constrain different objects by different mechanisms and agree; each is an exact, zero-parameter theorem.

§5.3 Exact reframing of gate FM13-5

Gate FM13-5 (ZS-M22, NC-M22.3) records that the additive transfer family W_p and the multiplicative gates M_p are not identified as operators. The flow picture gives this gap exact form. The same flow $U(t)$ admits two canonical prime samplings:

harmonic: $p \mapsto U(1/p) = W_p$, closure $\bigoplus_p \mathbb{Z}/p$ (torsion); logarithmic: $p \mapsto U(\log p)$, closure $\cong \bigoplus_p \mathbb{Z}$ (free).

Lemma 5.4 (logarithmic freeness). [PROVEN] If $\Sigma_p k_p \log p \in \mathbb{Z}$ with finitely many $k_p \in \mathbb{Z}$, then every $k_p = 0$ (and the integer is 0). *Proof.* Exponentiating, $\Pi_p p^{k_p} = e^m$ with $m \in \mathbb{Z}$. If $m \neq 0$, then e^m is transcendental by Lindemann's theorem [16] while the left side is rational — contradiction. If $m = 0$, unique factorization forces $k_p = 0$ for all p. □ Hence the logarithmic closure $\langle U(\log p)\rangle$ is free abelian of infinite rank. Check E4 is retained in the script only as a finite numeric witness; the proof above is exact — an audit-prompted upgrade of the v1.0 sample test, which by itself was no anti-numerology defence.

The FM13-5 gap is thus the gap between the torsion sampling (register side, detector) and the free sampling (locator side) of one flow — and the reason the free sampling cannot upgrade the register to a locator is Proposition 4.5: whatever times one samples, the generator **D** has eleven integer eigenvalues, and the trace remains the band-limited Dirichlet kernel with no arithmetic singular support. The gate is reframed exactly; it is not closed. [DERIVED-interpretation for the reframing; the constituent statements are PROVEN.]

§5.4 Finite Trace Rigidity (the exportable form of the boundary)

Theorem 5.5 (Finite Trace Rigidity). [PROVEN] Let H be a finite-dimensional Hilbert space, $D = D^\dagger$ with eigenvalues $\lambda_1, \ldots, \lambda_n$ ($n = \dim H$, with multiplicity), and $U(t) = e^{2\pi itD}$. Then $\mathrm{Tr}\, U(t) = \Sigma_j e^{2\pi i\lambda_j t}$, and as a tempered distribution its Fourier transform is $\Sigma_j \delta_{\lambda_j}$ — supported on at most n points. Consequently, for every generator D (integer spectrum or not) and every time-sampling set $\Gamma \subset \mathbb{R}$, the trace distribution of a finite-dimensional self-adjoint flow cannot possess the countably infinite singular support $\{\pm k \log p : p \text{ prime}, k \geq 1\}$, let alone the $\zeta_p(k)$-weighted measure, required of a Guinand–Weil locator by ZS-F25 Theorems 6.4-N/6.4-W. *Proof.* The spectral theorem gives the displayed sum; the Fourier transform of a finite sum of unimodular exponentials is the corresponding finite sum of Dirac masses; a set of cardinality ≤ n cannot contain an infinite set. □ (Verified: check T1 — a random integer-spectrum flow's trace has Fourier support exactly its distinct spectrum.) Finite rank alone, before any arithmetic enters, excludes locator status: this is the audit's Theorem candidate C, the paper's exportable, Z-Spin-free form of the detector/locator boundary.

Corollary 5.6. [DERIVED] Proposition 4.5 (the Dirichlet-kernel trace) and the §4.3 detector verdict are the instance $D = \mathrm{diag}(j - 5)$ of Theorem 5.5; and the Depth-One Obstruction (Theorem 5.1) and Theorem 5.5 are two faces of finiteness — a group-theoretic depth deficit and a support-theoretic rank deficit — either of which suffices for the exclusion. The corpus thereby holds the boundary in three independent forms: rank (ZS-F25 Lemma 6.5), depth (Theorem 5.1), and trace support (Theorem 5.5).

§6. The Dual Family and Register Weyl Kinematics

Definition 6.1. The Fourier-dual family is W_p^∨ := F W_p F^{-1} = e^{2πiD^∨/p}, with D^∨ = FDF^{-1} the register momentum operator.

Proposition 6.2. [PROVEN] Each W_p^∨ is a circulant operator (a polynomial in the shift S), since F conjugates the full diagonal algebra onto the full circulant (convolution) algebra; and [W_p, W_p^∨] ≠ 0 — the pair (W_p, W_p^∨) is a non-commuting Weyl-type clock–shift pair on the register. (Verified: checks F1–F4.) The intertwining F · (diagonal algebra) · F^{-1} = (convolution algebra) is the finite Fourier–Mukai equivalence — GL(1) geometric Langlands duality in dimension zero — exchanging the Wilson (spectral, order) side and the convolution (automorphic, disorder) side of the register.

§6.3 Compatibility with the abelian-by-foundation theorem

ZS-F25 §4 proves that the Wilczek–Zee holonomy of the pulsation attractor is abelian. The non-commutativity of (W_p, W_p^∨) does not contradict it and is not claimed to: the F25 theorem concerns the dynamical holonomy of the i-tetration flow, while the Weyl pair is operator kinematics on $\mathbb{C}^{11}$ — a space whose full operator algebra $\mathrm{Mat}_{11}(\mathbb{C})$ is trivially non-abelian. No claim is made that the corpus dynamics activates the dual family; whether any corpus-PROVEN structure (the Z-anchor vortex operators are the natural candidates, as disorder operators) is identified with W_p^∨ is registered OPEN (O-F27.1). [NON-CLAIM on holonomy non-abelianization.]

§6.4 Negative result: J is not the duality

Proposition 6.5. [DERIVED] The seam involution J maps the Wilson family to itself: J W_p J = W_p* (Corollary 3.5). It does not exchange the Wilson and convolution families; the operator that does is F (Proposition 6.2). In Kapustin–Witten language [7], J is a charge-conjugation-type involution within the electric family, not an S-duality; the register's duality operator is F. This negative identification, first reached in the internal deep-exploration session by confronting the candidate "J = S-duality shadow" with the already-PROVEN ZS-M25 identity, is registered here to prevent the overclaim.

§6.5 The seam generates the Weyl kinematics

Proposition 6.6 (generation). [PROVEN] (i) $S = F^2 J$ — immediate from Theorem 3.3, since $F^2 J = F^4 S = S$: the cyclic shift is recovered from the seam and the Fourier operator alone. (ii) The clock $M := FSF^{-1} = F^3 J F^3$ is diagonal. (iii) Hence $\langle F, J\rangle$ contains the full clock–shift pair, the group it generates contains the finite Heisenberg–Weyl group of $\mathbb{Z}_{11}$, and the *-algebra it generates is all of $M_{11}(\mathbb{C})$, with trivial commutant. (Verified: checks W1–W2.) This resolves the audit's Theorem candidate D in deflationary form: the seam–Fourier algebra admits no nontrivial classification because it is everything — the two canonical operators of this paper already generate the register's entire Weyl kinematics.

Incidence geometry. [COMPUTED] The refined object the audit asked for is therefore not the algebra but the relative position of the two canonical gradings — the seam grading $C\pm = \ker(J \mp I)$ (dimensions 6, 5) and the four Fourier chirality eigenspaces (dimensions 3, 3, 3, 2). At $N = 11$ they are in general position: all eight intersections $C_s \cap E_\lambda$ are zero, so F and J share no eigenvector — consistent with, and illustrative of, Proposition 6.6(iii). (Verified: check W3.) A closed-form incidence geometry (the principal-angle spectrum of the pairs $(P_{C\pm}, P_\lambda)$) for general odd N is registered as O-F27.6.

§7. The Tate Shadow

ZS-M25 (Appendix B, Category F) verifies, at PROVEN level, the intertwining $L_{1-s} = J \cdot L_s^\dagger \cdot J$ for the corpus Berry–Keating family — the register implementation of the functional-equation symmetry $s \leftrightarrow 1 - s$. Substituting Theorem 3.3:

$$L_{1-s} = (F^2 S) \cdot L_s^\dagger \cdot (S^{-1} F^2) .$$

Reading. [DERIVED-interpretation] The corpus functional-equation involution factors through the square of the register Fourier operator. In Tate's thesis [2], the functional equation of the completed zeta function is the statement of Fourier self-duality on the adeles — the $s \leftrightarrow 1 - s$ symmetry is generated by a Fourier transform. The register exhibits the same mechanism in finite dimension: its functional-equation symmetry is generated (up to the unit shift fixed uniquely by Theorem 3.4) by the square of a Fourier transform. We register this as a finite-dimensional shadow of the Tate mechanism, with three guards: the register F acts on $\mathbb{C}^{11}$, not on $\mathbb{A}_\mathbb{Q}$; the shadow is GL(1), depth-one, and torsion (§4–§5); and no statement about the actual functional equation of ζ is derived from it. The value of the shadow is structural: it shows that the corpus's central involution J and its prime-indexed family W_p are two projections of one canonical object — the register Fourier operator — exactly as Tate's thesis shows the classical functional equation and the local Euler factors to be two projections of adelic Fourier analysis.

§8. The Three-Component Decomposition of the Adelic Locator

§8.1 Condition L

Condition L (stated, not derived). The adelic locator — the operator-theoretic object whose spectrum is the nontrivial zeros, registered OPEN by ZS-F24 — admits a Langlands-type reading: it arises as the spectral side of a non-abelian correspondence over a (presently missing) global geometric object playing the role of a curve for $\mathbb{Q}$. Support for L within the corpus: ZS-F25 Theorem 6.3 (Twist test) confines all abelian twists to the Dirichlet family by Kronecker–Weber, so the locator requires GL(2)-and-above content; ZS-F25 Table 7.1 isolates the open quadrant (non-abelian, $h > 0$, adelic) as the locator's habitat; and the corpus's entire arithmetic apparatus (the characters χ_{-3}, χ_{-11}, χ_{33} and the V_4 Galois structure of ZS-M22/M25) is class field theory — abelian Langlands — exactly the GL(1) stratum whose boundary the Twist test proves. Condition L is nevertheless an interpretive commitment, not a theorem, and everything in this section is conditional on it.

§8.2 The decomposition

Theorem 8.2 (Three-Component Decomposition). [DERIVED-CONDITIONAL under L] Under Condition L, the adelic-locator OPEN of ZS-F24 decomposes as the conjunction of three components:

(A) **The global curve.** A geometric object over which "Spec $\mathbb{Z}$ becomes a curve" — the missing prerequisite. Status: OPEN at the external frontier. The leading candidates approach it from three directions: the Connes–Consani scaling site [9] (the corpus's own CCM interface, ZS-F24), the Deninger foliated flow [10] (the design occupant of the F25 open quadrant), and — at single primes — the Fargues–Fontaine curve, which already exists and carries component (B) locally [5].

(B) **The non-abelian spectral correspondence.** Status: PROVEN wherever a curve exists. Over smooth projective complex curves: the unramified geometric Langlands equivalence, Gaitsgory–Raskin et al. 2024 (GLC I–V) [3]. Over function-field curves: V. Lafforgue 2018 [4]. Over the Fargues–Fontaine curve (local, one prime): Fargues–Scholze 2021 [5]. Component (B) stood as a conjecture for three decades and was closed externally in 2024 — before the ZS-F24 → ZS-F26 arc was written; that arc, however, did not engage the Langlands column, and the closure was never registered in the corpus frame. Performing that registration — and stating exactly what it does and does not change for the locator OPEN — is the function of the present paper.

(C) **The unitarization layer.** The correspondence of (B) is categorical and carries no inner product; the locator is a self-adjoint spectral object. The unitary refinement — analytic Langlands — is proven in the Etingof–Frenkel–Kazhdan cases (compact self-adjoint commuting Hecke operators with real discrete spectrum, genus 0 with parabolic points over local fields) and open in general [6]. Status: partial.

Consequence. The locator OPEN is relocated, not closed: OPEN(locator) = OPEN(A) $\wedge$ PROVEN-where-curve(B) $\wedge$ partial(C). The compression is real — the conjunction has lost its middle conjecture — and the honesty constraint is equally real: components (A) and (C) suffice to keep the locator open, and nothing here advances RH. The structural echo with the Weil-positivity programme is registered at interpretation level: in ZS-F26 the explicit formula is PROVEN and the positivity layer is the frontier; in the Langlands column the correspondence is PROVEN (where geometric) and the unitarity layer is the frontier. Gate D5 and component (C) are the same kind of residue — the Hilbert-space layer of a structurally established correspondence. Provenance note (added in v1.1): the three-component narrative itself — a curve for Spec $\mathbb{Z}$, a non-abelian correspondence, a unitary spectral layer — is the long-standing consensus articulation of the cited programs [8–10]; what this section adds is its registration in the corpus frame under Condition L, with the 2024 status of (B) recorded. No external synthesis export analogous to the (H, h, P) triage of ZS-F25 is claimed.

§8.3 Extension of ZS-F25 Table 7.1

The F25 classification invariants are (H, h, P) = (holonomy class, entropy, place support). Two rows are appended; all prior rows are unchanged.

Table 8.1. Appended classification rows (extension of ZS-F25 Table 7.1).

Program	H	h	P	Classification predicts	Known status (published)
Gaitsgory–Raskin GLC (2024)	non-abelian (categorical)	— (no flow; categorical)	function-field / ℂ-curve (geometric)	full spectral decomposition where a curve exists; no ℚ-locator without component (A)	Unramified geometric Langlands PROVEN over complex curves (GLC I–V)
Etingof–Frenkel–Kazhdan analytic Langlands	non-abelian	— (operator family)	archimedean-analytic over local fields	unitary spectral layer: the (C)-component machine	Self-adjoint compact commuting Hecke operators, real discrete spectrum, PROVEN in genus-0 parabolic cases; general case open

The triage remains consistent: neither appended program occupies the open quadrant by itself — GLC lacks the flow/entropy and the ℚ-place; EFK supplies unitarity but not the global adelic substrate. The quadrant's design occupant (Deninger) and the curve candidates of component (A) are where the three columns must meet.

§9. Registration of the O-F26.5 Template

O-F26.5 (ZS-F26 §12) narrowed to the theoretical asymptotics of $\lambda_{min}(\ell)$ along the window ladder — the behavior of the minimal eigenvalue of the windowed Weil form as prime channels enter. The EFK mechanism — proving compactness of Hecke integral operators, hence discreteness and reality of the spectrum, before any explicit diagonalization — is registered as a closure-path template for O-F26.5: the windowed Weil operators of F26 are finite-window compressions of one operator (Connes's 2026 minimization reads its bottom; F26's MI(2) reads its positivity), and a compactness-style a priori control of the window family is the missing analytic ingredient that the EFK proofs supply in their setting. [HYPOTHESIS; explicit path: transport the EFK compactness scheme to the F26 half-window class of §3.1 of that paper.] No transport is performed here; the registration adds a frontier-synchronized route to the O-F26.5 ledger, parallel to the three routes already registered in ZS-F26 §7.7.

§10. Anti-Numerology Audit

Zero free parameters: every object of §3–§7 (F, S, P, D, J, W_p, U, $W_p^\vee$) is either corpus-LOCKED (Q = 11, J, W_p) or canonical given the register basis (F, S, D); no constant is chosen, fitted, or tuned. The audit registers four universality flags — places where a structure could be mistaken for a Q = 11 selection but is generic:

(i) F^2 = parity holds on $\mathbb{C}^N$ for all N (checked at N = 7, 12); Theorem 3.3's content is the identification of J, not a property of 11. (ii) The DFT eigenvalue multiplicity pattern for N = 11 is (3, 3, 3, 2) over {1, −1,

$\pm i\}$ by the universal McClellan–Parks count for $N \equiv 3 \pmod 4$ [1]; the fact that the deficient eigenspace has dimension $2 = \dim Z$, and that it is an eigenspace of eigenvalue $\pm i$ (the corpus's "one i"), is registered as OBSERVATION only, used in no proof — with the additional caution that which of $+i$ or $-i$ is deficient flips with the $\omega \leftrightarrow \bar{\omega}$ convention — under this paper's convention the deficient space is $\ker(F + iI)$. Correction of record: the v1.0 release shipped the convention-fixed form of this check, which fails under the paper's own convention; the released script therefore produced 27/28, not the 28/28 stated in v1.0. The discrepancy was found by independent external audit. Script v1.1 replaces the check by the convention-invariant pair H1a/H1b, adds the N-scan H2 (the witness of Proposition 3.7) and the odd-N seam witness C6, and reports 31/31. The incident is registered here and in the Version History. (iii) The Dirichlet-kernel trace (Proposition 4.5) is the generic trace of any integer-spectrum flow; its evidential content is the absence of arithmetic support, not the kernel's form. (iv) The depth-one closure is forced by the LOCKED definition $W_p = U(1/p)$; it is a discovered property of an existing corpus object, not a constructed coincidence. (v) Proposition 3.7 is the audit's most selection-sensitive item: the multiset statement is PROVEN, but the $\dim Z = 2$ link remains OBSERVATION, enters no proof, and cannot alone derive $Q = 11$ — other spectral patterns single out other N — so it bears on the corpus only jointly with the ZS-F5 decomposition. No Monte Carlo is required: the paper's claims are exact operator identities and conditional registrations, not numerical coincidences.

§11. Falsification Gates

Table 11.1. Falsification registry F-F27.1–F-F27.7.

Gate	Condition	Consequence if triggered	Class
F-F27.1	Any audit of corpus conventions under which $J \neq F^2S$ (indexing, basis, or sign).	§3–§7 collapse; immediate retraction of the seam factorization and all downstream readings.	Mathematical / immediate rejection
F-F27.2	$\mathrm{order}(W_p) \neq p$, or $W_p \neq U(1/p)$, for any prime under the LOCKED definition.	Flow reading (§4) fails; depth-one obstruction loses its premise.	Mathematical / immediate rejection
F-F27.3	The multiplicative closure of $\{W_p\}$ is found to contain an element of order p^2.	Depth-One Obstruction falsified — and a depth ≥ 2 register channel would thereby OPEN, a route this paper claims does not exist.	No-go gate (failure opens a route)
F-F27.4	A corpus-internal operator family is exhibited with trace singular support at $\{\pm k \log p\}$ and $\zeta_p(k)$ weights.	Proposition 4.5's detector verdict and Corollary 5.2 must be re-audited; the F24/F25 detector confinement is breached.	Structural / revision required
F-F27.5	The Gaitsgory–Raskin proof (GLC I–V) or V. Lafforgue's theorem is retracted or found flawed.	Component (B) of Theorem 8.2 reverts to OPEN; the decomposition survives but loses its 2024 compression.	External / dated revision
F-F27.6	EFK self-adjointness fails to extend in a way that breaks the compactness template.	The O-F26.5 route registration of §9 is withdrawn; F26's three prior routes are unaffected.	External / route withdrawal

F-F27.7	Condition L is shown untenable (a locator construction succeeds outside any Langlands-type reading).	Theorem 8.2 is voided as stated; §3–§7 (unconditional) are unaffected.	Conditional-frame collapse
F-F27.8	Independent execution of the released verification script yields any FAIL under the paper's stated conventions.	The corresponding claim reverts to unverified until script and paper are reconciled; precedent: the v1.0 H1 incident, corrected in v1.1.	Reproducibility / correction of record
F-F27.9	A nonzero seam-odd f on $\mathbb{Z}$_p with $\lvert\mathrm{supp}\, f\rvert\cdot\lvert\mathrm{supp}\, \hat{f}\rvert < 2(p - 1)$ is exhibited.	Theorem 3.9 retracted; the seam-constrained programme's first theorem falls.	Mathematical / immediate rejection
F-F27.10	A finite-dimensional self-adjoint flow whose trace distribution carries infinite singular support is exhibited.	Theorem 5.5 retracted (this would contradict the spectral theorem; the gate is registered for completeness of the multi-layer scheme).	Mathematical / immediate rejection
F-F27.11	Enumeration at any $N = p^n q$ exhibits a seam-invariant set that is spectral but does not tile, or tiles but is not spectral.	Corollary 3.12 and the §3.7 calibration re-audited (a confirmed instance would contradict the published theorem [22]).	Mathematical / immediate re-audit

§12. Open Problems and Non-Claims

O-F27.1. Corpus identification of the dual family: determine whether any corpus-PROVEN structure — the Z-anchor vortex operators (ZS-A6) being the natural disorder-side candidates — is identified with W_p^∨ = FW_pF^{-1}. A negative answer is itself registrable (the register's 't Hooft side would then be kinematically present but dynamically inert).

O-F27.2. Depth extension: U(1/p^2) exists on the flow but is not corpus-LOCKED. Determine whether any corpus principle selects the depth-k samplings U(1/p^k) — the register image of the F26 rung at k log p — or whether depth one is forced by a corpus axiom (in which case the Depth-One Obstruction would upgrade from a property of W_p to a theorem about the corpus).

O-F27.3. Derive or refute Condition L within the corpus: in particular, whether the F25 open-quadrant logic (Table 7.1) can be promoted from classification to necessity.

O-F27.4. Composite-N seam-odd uncertainty: extend Theorem 3.9 beyond prime N, where Tao's principle is unavailable and Meshulam-type divisor bounds [19] are expected to control the answer; the expected bound depends on the divisor structure of N.

O-F27.5 (= the ZS-F28 programme; narrowed in v1.3). The base family $\mathbb{Z}$_{$p^n q$} is closed by restriction of the import (Corollary 3.12; calibration only). The open target is the F28 main theorem: $N = p^n q^m$ ($p \neq q$ odd primes), A seam-invariant and spectral ⇒ A tiles, via the §3.7 skeleton; its open core is the Key Lemma (Lam–Leung decomposition under self-reciprocity [23]), whose viability is to be decided by structural enumeration before drafting.

O-F27.6. Closed-form seam–chirality incidence geometry: the principal-angle spectrum of the pairs (P_{C±}, P_λ) for general odd N, of which the N = 11 general-position fact of §6.5 is the first data point.

Non-claims. NC-F27.1: No claim on RH, GRH, or Weil positivity; Gate D5 remains IMPORTED-OPEN ≡ RH. NC-F27.2: No construction of, or claim about, the global curve of component (A); the scaling-site and foliated-flow candidates are cited at their published status. NC-F27.3: No claim that geometric Langlands applies to $\mathbb{Q}$ today. NC-F27.4: No claim that the register Weyl pair non-abelianizes the corpus dynamics or holonomy (§6.3). NC-F27.5: The Tate shadow (§7) asserts a structural factorization on $\mathbb{C}^{11}$ only; nothing about ζ's functional equation is derived from it. NC-F27.6: The OBSERVATION of §10(ii) (deficient ±i eigenspace of dimension 2 = dim Z) carries no evidentiary weight. NC-F27.7: The seam-restricted Fuglede programme is not claimed closed; Corollary 3.12 is a calibration closure of an imported, already-settled base family — explicitly not a new theorem — and §3.7 contributes no new external mathematics beyond Lemma 3.13; the center-slot coincidence at N = 11 is OBSERVATION only.

§13. Conclusion

ZS-F27 adds one internal floor and one external ceiling to the honest terminus of the RH-bridge arc. The internal floor is exact and unconditional: the register $\mathbb{C}^{11}$ carries a canonical Fourier–Langlands structure of which the corpus's central involution and its prime-indexed transfer family are projections — $J = F^2S$ (Theorem 3.3, with the Canonical Seam–Fourier Theorem 3.4 fixing its uniqueness at every odd N), $W_p = U(1/p)$ on one flow (Proposition 4.1), closure $\bigoplus_p \mathbb{Z}/p$ with dual $\prod_p \mathbb{F}_p$ (Theorem 4.2, Corollary 4.3), the Dirichlet-kernel trace (Proposition 4.5), the circulant dual family and Weyl kinematics (§6), and the Tate-shadow factorization of the functional-equation involution (§7). From it follows the Depth-One Obstruction (Theorem 5.1), the second independent finite-versus-infinite exclusion in the corpus after the F25 rank obstruction, and the exact reframing of gate FM13-5 as the torsion/free dichotomy of two prime samplings of one flow. The external ceiling is the Three-Component Decomposition (Theorem 8.2, conditional on the stated Condition L): the locator OPEN now reads curve (OPEN) ∧ correspondence (PROVEN where geometric, by the 2024 closure of geometric Langlands) ∧ unitarization (partial) — a relocation that removes the middle conjecture from the conjunction while leaving the conjunction open. The register structure and the frontier decomposition are the same picture at two scales: the register realizes, exactly and provably, the GL(1), depth-one, torsion, archimedean floor of the tower whose non-abelian, all-depth, free, adelic ceiling is the locator. What separates floor from ceiling is now stated with theorem-grade precision — and nothing in that statement is, or is claimed to be, a step onto the ceiling. Version 1.2 adds the first steps outward rather than upward: the seam, taken not as a label but as a symmetry constraint, doubles the prime uncertainty product (Theorem 3.9), pins the Fourier phases of invariant masks to the half-angle line (Lemma 3.11), generates the entire Weyl kinematics together with F (Proposition 6.6), and — in its exportable form — yields the finite trace rigidity no-go (Theorem 5.5); the seam-constrained programme these seed is designated ZS-F28 — which version 1.3 calibrates on its first imported base family (Corollary 3.12, with exhaustive non-vacuity at N = 15) and points at its main target, $N = p^n q^m$.

Acknowledgements & Code Availability

This work consolidates internal Z-Spin Collaboration research notes from the Geometric Langlands × Z-Spin deep-exploration sessions (five-summit survey; Gaitsgory–Raskin engagement; flow-curve session; paper-architecture session). All numerical checks — the seam factorization and its uniqueness,

the Wilson flow and closure structure, the dual family and Weyl non-commutativity, the Dirichlet-kernel trace, the DFT eigenvalue multiplicities including the documented convention-dependence correction — are reproduced by the verification script zs_f27_verify_v1_3.py (Python/NumPy, 42 checks, machine precision), available at https://github.com/KennyKang-git/zspin/tree/main/verify_scripts. Expected output: 42/42 PASS, exit code 0. Version 1.1 incorporated two independent audit reports on v1.0: a verification-reproduction audit, which found the H1 incident (§10), and an external-value assessment, which prompted the genre statement of §1.1, the theorem promotions of §3, and the provenance note of §8. Version 1.2 responds to a third audit report, which proposed the seam-constrained finite Fourier programme and four theorem candidates; their disposition is: candidate A proved (Theorem 3.9), candidate C proved (Theorem 5.5), candidate B reduced to its provable kernel (Lemma 3.11) plus the registered ZS-F28 programme (O-F27.5), and candidate D resolved deflationarily (Proposition 6.6) with its refined object registered open (O-F27.6). Version 1.3 responds to a fourth audit report, which prescribed the pilot-closure scope (calibration only in F27; the novelty-bearing theorem separated into ZS-F28) and the prohibited-overclaim phrasing adopted in §3.7 and NC-F27.7.

Appendix A. Verification Table

Table A.1. The 42 verification checks (zs_f27_verify_v1_3.py).

No.	Check	Result
A1	$Q = 11$ register, basis $|0\rangle\ldots|10\rangle$ (LOCKED input)	PASS
A2	$J^2 = I$ (corpus involution)	PASS
A3	$S^{11} = I$ (cyclic shift)	PASS
B1	F unitary	PASS
B2	F^2 = parity $|j\rangle \to |-j \bmod 11\rangle$	PASS
B3	$F^4 = I$	PASS
C1	$J = F^2S$	PASS
C2	$J = S^{-1}F^2$	PASS
C3	$F^2SF^2 = S^{-1}$	PASS
C4	$T_a D T_a^{-1} = -D$ iff $a = 1$ (uniqueness over all 11 translates)	PASS
C5	$JDJ = -D$	PASS
C6	Unique $a = 1$ reverses centered D for all odd $N \in \{5, 7, 9, 11, 13\}$ (Theorem 3.4 witness)	PASS
D1	$J W_p J = W_p^*$ for $p \in \{2, 3, 5, 7, 13\}$ (ZS-M25 consistency)	PASS
D2	$W_p = U(1/p)$ for $p \in \{2, 3, 5, 7, 11, 13\}$	PASS
D3	$\mathrm{order}(W_p) = p$ exactly	PASS
D4	$[W_p, W_q] = 0$ (full sample grid)	PASS
E1	$U(t) = I$ iff $t \in \mathbb{Z}$ (sample $t = 1/2, 1/4, 1$)	PASS
E2	$|\langle W_2, W_3, W_5\rangle| = 30 = 2\cdot3\cdot5$ (depth-one product)	PASS

E3	1/4 (depth two of $p = 2$) not in closure sample	PASS
E4	Numeric witness for Lemma 5.4 (logarithmic freeness; proof exact via unique factorization + Lindemann [16])	PASS
F1	$W_p^\vee = FW_pF^{-1}$ circulant ($p = 2, 3, 7$)	PASS
F2	$W_p^\vee = \exp(2\pi i D^\vee/p)$, $D^\vee = FDF^{-1}$	PASS
F3	$[W_p, W_p^\vee] \neq 0$ (Weyl-type pair)	PASS
F4	$F\cdot$(diagonal algebra)$\cdot F^{-1}$ = circulant algebra (random diagonal test)	PASS
G1	$\mathrm{Tr}\, U(t) = \sin(11\pi t)/\sin(\pi t)$ at four sample times	PASS
G2	Fourier support of register trace = $\{-5, \ldots, 5\}$ (exact, by construction)	PASS
H1a	Multiset (3, 3, 3, 2); deficient eigenvalue $\in \{\pm i\}$, eigenspace dim 2 (ω convention: $-i$) — replaces the v1.0 convention-fixed H1, which FAILED in release (§10(ii))	PASS
H1b	Same multiset under the conjugate convention; deficient chirality flips to $+i$ (convention invariance)	PASS
H2	Multiset (3, 3, 3, 2) occurs for $2 \leq N \leq 60$ iff $N = 11$ (Proposition 3.7 witness)	PASS
I1	Zero free parameters (all objects LOCKED or canonical)	PASS
I2	Universality flag: F^2 = parity at $N = 7, 12$ (not $Q = 11$-selective)	PASS
U1	Seam-odd structure lemma: center = 0, even supports, $\hat{f}(0) = 0$, dual relation $\hat{f}(-m) = -\omega^m \hat{f}(m)$ ($p = 11$)	PASS
U2	min $\lvert\mathrm{supp}\, f\rvert\cdot\lvert\mathrm{supp}\, \hat{f}\rvert$ over exhaustive seam-pair patterns × random coefficients = $2(p - 1)$ exactly, $p \in \{5, 7, 11\}$	PASS
U3	Equality classification witness: every seam dipole gives $(2, p-1)$; every dual dipole gives $(p-1, 2)$ and is seam-odd ($p = 11$)	PASS
P1	Seam-invariant mask $\Rightarrow \zeta^{-(N-1)/2}A(\zeta) \in \mathbb{R}$, random sets, $N = 11, 15$ (Lemma 3.11)	PASS
T1	Random integer-spectrum flow: trace Fourier support = distinct spectrum ($\leq$ dim); no logarithmic frequencies (Theorem 5.5)	PASS
W1	$S = F^2J$ (shift recovered from seam + Fourier)	PASS
W2	Clock $M = FSF^{-1} = F^3JF^3$, diagonal (Proposition 6.6)	PASS
W3	Seam grading and Fourier chirality in general position at $N = 11$: all eight intersections zero	PASS
S1	Exhaustive $N = 15$: spectral $\Leftrightarrow$ tile over all 255 non-empty seam-invariant subsets (16 = 16, zero mismatches; Corollary 3.12 base witness)	PASS
S2	Non-vacuity: $\{0, 7, 14\}$ is a seam-invariant non-coset tile; every seam-invariant tile contains the center slot	PASS
Q1	Seam quotient stability: $\pi(A)$ seam-invariant for all 256 seam-invariant $A \subseteq \mathbb{Z}_{15}$ under $\mathbb{Z}_{15} \to \mathbb{Z}_5, \mathbb{Z}_3$ (Lemma 3.13)	PASS

Appendix B. Closure Computation for Theorem 4.2(iii)

The subgroup of $\mathbb{Q}/\mathbb{Z}$ generated by $\{1/p : p \text{ prime}\}$ consists of classes $t = \Sigma_p\, k_p/p \bmod 1$ with finitely many $k_p \in \mathbb{Z}$. The p-primary component of $\mathbb{Q}/\mathbb{Z}$ is $\mathbb{Z}[1/p]/\mathbb{Z} \cong \lim \mathbb{Z}/p^k$ (Prüfer group); the image of the generated subgroup in this component is $(1/p)\mathbb{Z}/\mathbb{Z} \cong \mathbb{Z}/p$, since cross terms k_q/q ($q \neq p$) contribute only prime-to-p denominators. Hence the closure is $\bigoplus_p \mathbb{Z}/p$, no element has order p^2, and the Pontryagin dual is $\prod_p \mathbb{Z}/p = \prod_p \mathbb{F}_p$. The finite witness computed in check E2 — the subgroup generated by 1/2, 1/3, 1/5 has order exactly 30 — and the depth witness E3 ($1/4 \notin$ closure) instantiate the statement; the proof above is exact and independent of the witnesses. □

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Version History

v1.0 (March 2026): Initial public release. Seam Factorization J = F²S with uniqueness lemma (§3, PROVEN); Wilson flow W_p = U(1/p), closure ⨁_p ℤ/p, Pontryagin dual ∏_p 𝔽_p, Dirichlet-kernel trace (§4, PROVEN); Depth-One Obstruction complementing the ZS-F25 rank obstruction and exact reframing of gate FM13-5 (§5, DERIVED); dual family W_p^∨, register Weyl kinematics, and the negative identification J ≠ S-duality (§6); Tate-shadow factorization of L_{1−s} = J·L_s†·J (§7, DERIVED-interpretation); Three-Component Decomposition of the adelic locator under Condition L, importing Gaitsgory–Raskin 2024, V. Lafforgue 2018, Fargues–Scholze 2021, and Etingof–Frenkel–Kazhdan (§8, DERIVED-CONDITIONAL); ZS-F25 Table 7.1 extended by two rows; EFK compactness template registered to O-F26.5 (§9, HYPOTHESIS); anti-numerology audit with four universality flags including the documented H1 convention-dependence correction (§10); falsification gates F-F27.1–F-F27.7; open problems O-F27.1–O-F27.3. Verification claimed 28/28 PASS (corrected in v1.1: the released v1.0 script produced 27/28; see the v1.1 entry). Zero free parameters. (A, Q, dim Z) = (35/437, 11, 2) LOCKED unchanged. No claim on RH. (Consolidated from internal Z-Spin Collaboration research notes up to the Geometric Langlands × Z-Spin deep-exploration sessions, v1.0.0 → v1.0.)

v1.1 (March 2026): Correction and sharpening release following two independent audits of v1.0. (1) Correction of record: the v1.0 verification script shipped with a convention-fixed eigenstructure check (H1) that fails under the paper's own ω convention; the released script produced 27/28 while the paper stated 28/28. Script v1.1 replaces H1 by the convention-invariant pair H1a/H1b, adds the N-scan H2 and the all-odd-N seam witness C6, and reports 31/31 PASS; the incident is documented in §10(ii) and gate F-F27.8 is added. (2) Lemma 3.4 promoted to Theorem 3.4 (Canonical Seam–Fourier Theorem), stated and proved for every odd N. (3) New §3.4 with Proposition 3.7: the DFT spectral multiset (3, 3, 3, 2) occurs at N = 11 and at no other N (McClellan–Parks count; scan witness H2); the dim Z = 2 link is held at OBSERVATION with an explicit non-derivation guard. (4) The v1.0 sample check E4 upgraded to Lemma 5.4 (logarithmic freeness), proved exactly via unique factorization and Lindemann's theorem (new import [16]). (5) Genre statement added (§1.1): operator-normalization and registration paper, no external-novelty claim; provenance note added to §8; Abstract, metadata, Acknowledgements, and Appendix A updated accordingly. No mathematical statement of v1.0 is retracted; the v1.0 verification claim is corrected as stated.

v1.2 (March 2026): Programme-opening release following a third audit, which proposed seam-constrained finite Fourier analysis (impose $J_N = F^2S$ as a symmetry constraint; prove external theorems first, reinterpret after) and four theorem candidates. Disposition: (A) PROVEN — Seam-Odd Uncertainty Theorem on prime cyclic groups, $|\mathrm{supp}\, f|\cdot|\mathrm{supp}\, \hat{f}| \geq 2(p - 1)$ for nonzero seam-odd f, with complete equality classification (dipoles and dual dipoles), via the structure Lemma 3.8 and Tao's prime uncertainty principle (new import [17]); new §3.5, checks U1–U3. (C) PROVEN — Finite Trace Rigidity Theorem 5.5: no finite-dimensional self-adjoint flow, under any time sampling, can produce a Guinand–Weil-type logarithmic singular support; Proposition 4.5 and the Depth-One Obstruction become instances (Corollary 5.6); new §5.4, check T1. (B) reduced to its provable kernel — half-angle phase pinning of seam-invariant masks (Lemma 3.11, check P1) — with the seam-restricted Fuglede programme registered HYPOTHESIS-strong as O-F27.5 and designated ZS-F28 (new imports [20–22]). (D) resolved deflationarily — $S = F^2J$ and $M = F^3JF^3$ (Proposition 6.6, checks W1–W2): the seam and the Fourier operator generate the full Weyl kinematics, so the algebra admits no nontrivial classification; the refined incidence geometry is COMPUTED at N = 11 (general position, check W3) and registered O-F27.6. Gates F-F27.9–10 added; O-F27.4–6 added; §1.1 genre statement amended (first external-grade theorems, stated Z-Spin-free); references [17–22] added; verification 39/39 PASS (script v1.2). All v1.0/v1.1 statements unchanged.

v1.3 (March 2026): Pilot-closure release following a fourth audit, which prescribed scope discipline: calibration only in F27, the novelty-bearing theorem separated into ZS-F28. New §3.7: Corollary 3.12 — on $\mathbb{Z}_{p^n q}$ the seam-invariant Fuglede equivalence holds as the restriction of the imported Malikiosis–Kolountzakis theorem [22] (DERIVED from IMPORTED-PROVEN; explicitly not a new theorem; NC-F27.7 added); Lemma 3.13 — seam-invariance is stable under every quotient (PROVEN; the structural prerequisite of the F28 induction, supplied because the audit's induction step presupposed it); non-vacuity established by exhaustive enumeration at N = 15 (255 non-empty seam-invariant subsets; 16 spectral = 16 tiles, coinciding set-for-set; non-coset examples; every seam-invariant tile contains the center slot, with the N = 11 kinematic-fixed-point coincidence held at OBSERVATION); ZS-F28 main target formally designated ($N = p^n q^m$, seam-invariant spectral ⇒ tile; the converse expected via Coven–Meyerowitz/Łaba, to be confirmed at drafting) with a five-step skeleton whose open core is the Key Lemma (Lam–Leung under self-reciprocity; new imports [23–24]); O-F27.5 narrowed; gate F-F27.11 added; verification 42/42 PASS (script v1.3, checks S1–S2, Q1). All earlier statements unchanged.